

- Q.18 Define refrigeration and air conditioning. Also write at least two applications of refrigeration and air conditioning in agriculture.
- Q.19 Discuss various control and prevention of corrosion.
- Q.20 Define elasticity and plasticity with example.
- Q.21 Enlist various physical properties of metals.
- Q.22 Describe the concepts of water management.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Describe hydrologic cycle & labelling their parts names with neat and clean sketch.
- Q.24 Write down various mechanical properties of metals. Explain any four of them.
- Q.25 a) Define various modes of heat transfer.
b) Define dumpy level, its uses and settings.

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Roll No.

1st Year / Agri Tech.

Subject : Basics of Agricultural Engineering

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 Which branch of surveying is used to find the elevations of given points with respect to given or assumed datum?
- Contouring
 - Traversing
 - Levelling
 - Plane table surveying
- Q.2 The ability of a material to withstand wear and abrasion is known as :
- Hardness
 - Fatigue Strength
 - Toughness
 - Wear Resistance

- Q.3 The ratio of specific weight of a liquid to the specific weight of pure water at a standard temperature is called
- Density of liquid
 - Compressibility of liquid
 - Specific gravity of liquid
 - Surface tension of liquid
- Q.4 _____ is the process where plants lose water, primarily in the form of vapour, from their aerial parts, mainly through tiny pores.
- Evapotranspiration
 - Transpiration
 - Precipitation
 - Runoff
- Q.5 The agricultural practice of planting seeds in the soil, with the goal of cultivating them into mature plants is called
- Tillage
 - Harvesting
 - Threshing
 - Sowing
- Q.6 Heat transfer takes place according to which of the following law?
- Second law of thermodynamics
 - First law of thermodynamics
 - Newton's second law of motion
 - Newton's law of cooling

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SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 Removal of unwanted plants from main crop is called_____.
- Q.8 Levelling deals with measurements in a _____.
- Q.9 Define drainage.
- Q.10 Define corrosion.
- Q.11 Define runoff.
- Q.12 Write SI unit of thermal conductivity.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Write down various resources used in agriculture.
- Q.14 List down any four field operations. And also write down machinery used for those operations in agricultural farm.
- Q.15 What are contour intervals? Also write any two factors affecting contour intervals.
- Q.16 Enlist various pressure measuring devices.
- Q.17 Discuss various factors affecting heat energy.

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